21st CENTURY COURSE OUTLINE FOR 2026/2027.

UNIVERSITY OF CAPE COAST

DEPARTMENT OF PHYSICS

FIRST SEMESTER-2026/2027 ACADEMIC YEAR

Course Code /Title: PHY 101 GENERAL PHYSICS 1 (THEORY)

Credit: (2 hours of lecture)

Lecturer: Mr. Yusif Owusu

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Office Hours: As announced in class

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Meeting hours: See Time table – Any Extra Meeting Hour Needed (To be discussed in class)

Course Description

This course provides an introduction to general physics and in mechanics. The content deals with:

Introduction to mechanics, vectors, dynamics, kinematics, work and energy, elasticity, fluids and thermodynamics.

Course Objectives:

At the end of the course, the student should be able to:

- Identify physical quantities and standard units, deduce derived quantities from the fundamentals and convert other units to SI units.
- Differentiate between vectors and scalar quantities, identify components of vector, and apply mathematical operators like addition, subtraction and multiplication on vectors.
- Define force, discuss types of forces, discuss Newton's laws of motion, explain impulse and momentum, discuss inelastic and elastic collision, strings and pulleys, inclined planes.
- Explain displacement, speed, velocity and acceleration, plot graphs of motion, and discuss the following types of motion :(Circular motion, simple harmonic motion and projectile motion).
- Explain work, energy and power, derive expressions for work, kinetic energy, potential energy and power.
- Explain elasticity, define Hooke's law and describe an experiment to verify Hooke's law, discuss young's modulus, shear modulus and bulk modulus.

- Explain the state of matter, explain density and relative density, discuss pressure in liquids, describe an experiment to verify Pascal's law and explain buoyancy and Archimedes principle.
- Explain kinetic theory of gases, discuss temperature scales, explain lab and clinical thermometers, discuss heat capacities of matter and explain thermal conductivity.

Mode of Teaching:

The course is taught in two modes via face-to-face classroom setting and E-learning platform with relevant teaching (visual and audio-visual) aids to enhance students' understanding of the concepts. Lectures are done in the form of discussion where students are allowed to express themselves in other to bring divers ideas about the topic under consideration which promotes intellectual flexibility. For each group presentation, a leader is chosen by the members of the team and the rest support him/her. This help individual students to improve their leadership skills. At end of every major topic, students are given trial questions to solve to enable them to reflect on what have been discussed.

Mode of Assessment: Two (2) quizzes which carry 10 marks each, Assignments (10 marks) and Power point presentations (10 marks). The created groups of students are given different topics to analyze critically and present in class (Power point presentation).

Exams: The exams (60%) consist of qualitative and quantitative problems similar to those discussed in class and those assigned for homework.

WEEK	TOPICS	OBJECTIVES	ACTIVITY	CORE VALUES
1	**Introduction to mechanics** ➢ Fundamental and Derived Quantities, Standard Prefixes, Unit Conversions ➢ Dimensional Analysis	1. Identify physical quantities and standard units. 2. Deduce derived quantities from the fundamentals. 3. Write figures in standard forms and prefixes. 4. Convert other units to SI units. 5. Apply dimension analysis on some equations.	Discussion of physical quantities and their SI units. Using fundamental quantities to obtain SI units of speed, force, volume, etc. Apply dimension analysis on pressure, area, volume, density, etc.	Equal opportunities and brainstorming enhanced. Identify and derive.
2 & 3	**Scalars and vectors** ➢ Components of a vector ➢ Unit vectors ➢ Addition and subtraction of vectors	1. Identify scalar and vector quantities. 2. Identify components of vectors.	Discussion on scalar and vector quantities with examples. Identifying components of	Reason Solve Identify Empowerment Quiz 1

	➢ Multiplication of vectors	3. Apply mathematical operators like addition, subtraction and multiplication on vectors.	vectors, calculating their magnitudes, directions and unit vectors. Solve problems on vectors.	
4 & 5	**Dynamics** ➢ Force and types of forces ➢ Newton's laws of motion ➢ Impulse and momentum ➢ Inelastic and elastic collisions ➢ Strings, pulleys and inclined planes	1. Define force. 2. Discuss types of forces. 3. Discuss Newton's laws of motion. 4. Explain impulse and momentum. 5. Discuss inelastic and elastic collision, strings and pulleys, inclined planes.	Lecture/ Discussion on forces and types of forces. Power point presentation by groups on Newton's law of motion. Discussion and solving questions on elastic and inelastic collisions, strings, pulleys and inclined plane.	Assignment in groups. Team work Leadership skills Use of ICT Equal opportunity and empowerment
6 & 7	**Kinematics** ➢ Displacement, speed, velocity and acceleration ➢ Graphs of motion ➢ Circular motion ➢ Simple harmonic motion ➢ Projectile motion	1. Explain displacement, speed, velocity and acceleration. 2. Plot graphs of motion. 3. Discuss the following types of motion: i. Circular motion ii. Simple harmonic motion. iii. Projectile motion	Lecture/ Discussion on displacement, speed, velocity and acceleration. Plotting displacement-time graph, velocity- time graph, and acceleration-time graph. Discuss and solve problems on circular motion, simple harmonic motion and projectile motion. Students are given	Assignment in groups. Team work Leadership skills Use of ICT Equal opportunity Quiz 2

			assignment on the different types of motions.	
8	**Work, Energy and Power**	1. Explain work, energy and power. 2. Derive expressions for work, kinetic energy, potential energy and power. 3. Solve questions.	Lecture/ Discussion on work energy and power. Guiding students to solve problems on work, energy and power. Students are given problems to solve.	Assignment in a groups. Team work Leadership skills Use of ICT Equal opportunity
9 & 10	**Elasticity** ➢ Stress and string ➢ Young's modulus ➢ Shear modulus ➢ Bulk modulus ➢ Hooke's law	1. Explain elasticity. 2. Define Hooke's law. 3. Describe an experiment to verify Hooke's law. 4. Discuss young's modulus, shear modulus and bulk modulus. 5. Solve problems.	Lecture/ Discussion on elasticity and Hooke's law. Describing an experiment to verify Hooke's law Power point presentation by groups on young's modulus, shear modulus and bulk modulus.	Assignment in groups. Team work Leadership skills Use of ICT Equal opportunity and empowerment.
11	**Fluids** ➢ State of matter (liquid and gas) ➢ Density and Relative density ➢ Pressure in liquids ➢ Pascal's law ➢ Buoyancy and Archimedes principle	1. Explain the state of matter. 2. Explain density and relative density. 3. Discuss pressure in liquids. 4. Describe an experiment to verify Pascal's law. 5. Explain buoyancy and Archimedes principle.	Lecture/ Discussion on state of matter. Explaining density and relative density. Discussion on pressure in liquids. Describing an experiment to verify Pascal's law. Discussing on buoyancy and	Reason Solve Identify Empowerment Team work

			Archimedes principle	
12	**Heat energy** ➢ Kinetic theory of gases ➢ Temperature scales ➢ Lab and clinical thermometers compared ➢ Heat capacities of matter ➢ Thermal conductivity	1. Explain kinetic theory of gases. 2. Discuss temperature scales. 3. Explain lab and clinical thermometers. 4. Discuss heat capacities of matter. 5. Explain thermal conductivity.	Lecture/ Discussion on kinetic theory of gases and temperature scales. Discussion on lab and clinical thermometer. Power point presentation by groups on heat capacities and thermal conductivity.	Assignment in groups. Team work Leadership skills Use of ICT Equal opportunity and empowerment.
13	**Revision**			

Tutorial sessions: 1 hour per week where Quizzes and Assignments are thoroughly discussed.

Reference

1. Halliday, D., Resnick, R., & Walker, J. (2021). *Fundamentals of Physics* (12th ed.). John Wiley & Sons.
2. Serway, R. A., & Jewett, J. W. (2018). *Physics for scientists and engineers*. Cengage learning
3. Cutnell, J. D., & Johnson, K. W. (2014). *Physics, Volume One: Chapters 1-17* (Vol. 1). John Wiley & Sons.